

Arun Kumar

6387681461 | arunk02615@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Computer Science Engineering student with strong problem-solving skills in **Data Structures & Algorithms** and a passion for building scalable software solutions. Experienced in **full-stack development** using the MERN stack, with hands-on exposure to **integrating LLMs and GenAI** capabilities into real-world applications. Constantly exploring the intersection of core CS fundamentals and applied AI to build efficient, intelligent products.

EDUCATION

Rajkiya Engineering College Sonbhadra

BTech, Computer Science and Engineering, CGPA: 7.5

Sonbhadra, UP

Sep. 2023 – Aug 2027

WCA Inter College

Class X: 90.33% | Class XII: 83.8%

Fatehpur, UP

Aug. 2019 – Jul 2022

TECHNICAL SKILLS

Languages: C/C++, JavaScript, Python, SQL, HTML5, CSS

Web Dev: ReactJS, NodeJS, FastAPI ExpressJS, Tailwind CSS, RESTful API, Socket.io, MongoDB, Authentication

Developer Tools: Git, GitHub, VS Code, Postman API, Cloudinary, Render, Vercel

AI/ML: LangChain, LLMs, RAG, Hugging Face, LangGraph, Tools, Vector DB, Gemini API, Agents

Computer Fundamentals: Data Structures & Algorithm, OOPs, Database Management System, Computer Networks, Operating System

EXPERIENCE

Infosys Springboard

AI Intern

Aug 2026 – Present

Remote

- Developed machine learning models for solving real-world problems using Python and Scikit-learn.
- Preprocessed and analyzed datasets using Pandas and NumPy to improve model accuracy.
- Implemented NLP techniques for text analysis and feature extraction.

PROJECTS

AskPDF - RAG Based Q&A system | *NodeJs, Redis, MongoDB, Gemini API*

CODE

- Engineered an end-to-end **Retrieval-Augmented Generation (RAG)** platform that allows users to interact with uploaded PDFs via conversational **LLMs**, featuring ultra-low-latency **real-time response streaming**.
- Architected a robust **asynchronous ingestion engine** (parsing → text chunking → embedding generation → vector indexing → search) capable of processing large PDF files up to **20MB** using **768-dimensional vector** embeddings.
- Offloaded PDF processing to a **decoupled Redis Worker** using a **BRPOP** blocking queue, enabling the API to return **202 Accepted** immediately while tracking asynchronous job status via **Redis Hashes (HSET)**.
- Implemented strict context filtering algorithms with **similarity score cutoffs** (≥ 0.7) to **eliminate hallucinations**, alongside automated **cleanup mechanisms** to prevent orphaned data in MongoDB during task failures.

AI Code Review Tool | *ReactJs, NodeJs, ExpressJs, MongoDB, Groq AI, JWT*

CODE | LIVE

- Built a full-stack AI-powered code assistant using **MERN stack**, integrating **Groq's LLaMA 3.3** model for real-time code review, bug detection, and explanation via a **ChatGPT-like** chat interface.
- Implemented **persistent chat history** using MongoDB — users can resume previous conversations anytime, and designed **account-based rate limiting** restricting 50 messages per 4-hour window to prevent API abuse.
- Secured all endpoints with **JWT authentication** and **bcrypt password hashing** — deployed frontend on **Vercel** and backend on **Render** with **MongoDB Atlas** as cloud database.

Blogify | *Node.js, Express.js, MongoDB, EJS, JavaScript, Cloudinary, JWT*

CODE | LIVE

- Developed a full-stack blogging application using **Node.js**, **Express.js**, and **MongoDB**, enabling users to create, manage, and publish blog posts.
- Implemented **JWT-based authentication** and **bcrypt password hashing** to enhance application security and protect user credentials.
- Designed dynamic server-side rendered pages using **EJS** and integrated **MongoDB** with Mongoose for efficient data storage and retrieval.

ACHIEVEMENTS

- Solved **1250+ DSA problems** across renowned platforms including LeetCode, CodeChef, Codeforces and GeeksforGeeks, demonstrating strong problem-solving skills.
- Achieved a **Global Rank 2402** among **27350** participants in LeetCode Weekly Contest 501, also secured **Global Rank 1352** in CodeChef Starters 224, competing with international participants.
- Secured **Rank 295** in Round 1 and **Rank 732** in Round 2 in CodeClash, a coding contest organized by the Let's Code Community, competing among thousands of participants.
- Cleared **The Big Code Contest** organized by **Google**, securing a position among the top 15,000 out of 3,00,000+ participants worldwide.
- Coding Profiles:** LeetCode, GeeksforGeeks